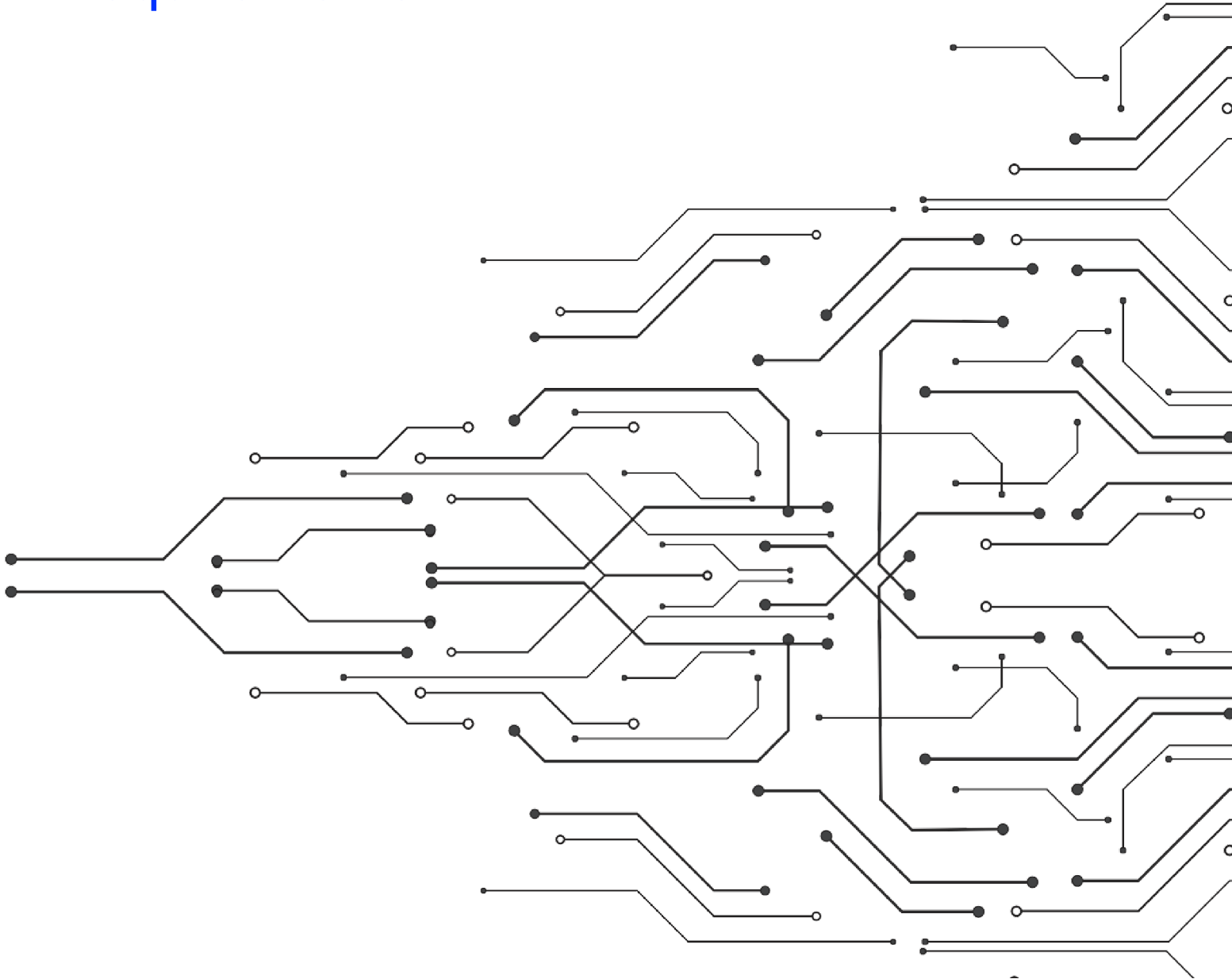




TEAM CHALLENGE

Requirements



CHALLENGE STATEMENT

GIC's complements its investment strategy by investing in funds from external providers (e.g. Blackrock, Bridgewater). By investing in these funds and related products, asset managers can gain broad diversification across alpha sources, and access to risk drivers with low correlations to core holdings. Funds can also provide exposure to differentiated strategies or new geographies. While this allows GIC to tap onto the expertise of institutional firms world-wide, it also increases the complexity of how GIC manages risk, measure performance and comply with cross-border regulations. Details of the investment positions are held with the external providers, but they can provide these investment details to GIC as CSV data files, or through API endpoints.

As such, GIC needs your team to develop a platform to ingest these investment details and provide richer "look-through" analytics for the underlying securities held in these funds. The platform will also need to standardize information from multiple source parties into GIC's common taxonomy, to bring meaningful analytics to investment risk professionals.

Your team of software developers believe the team can achieve this with modern technologies and would love to develop a well-engineered solution to achieve new look-through capabilities for senior management.

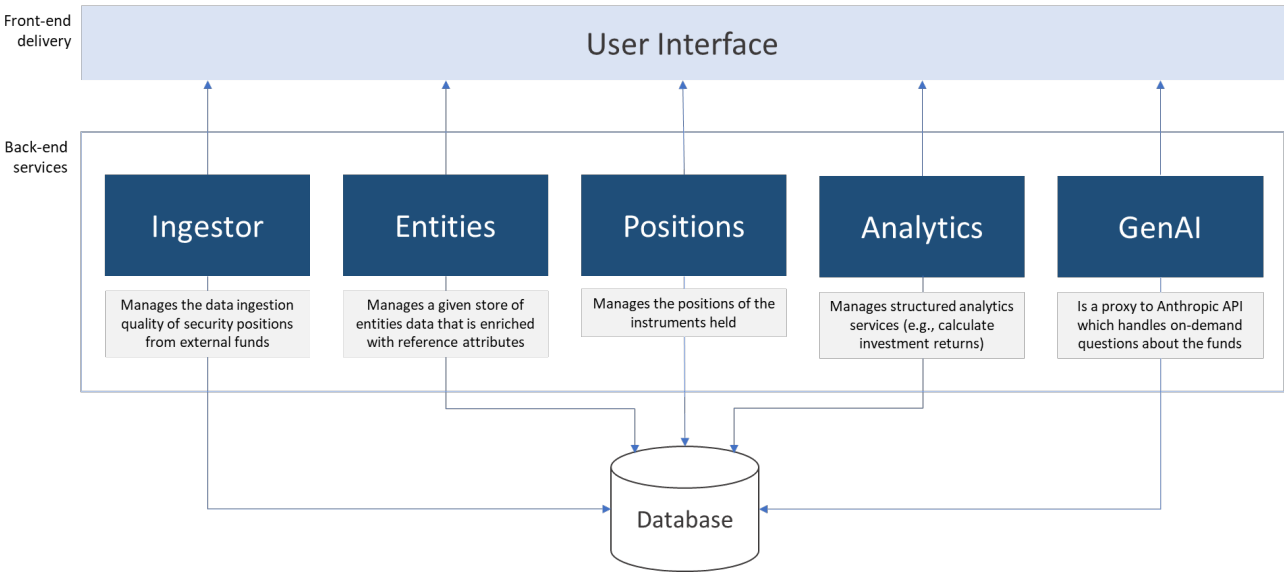
TECHNICAL OVERVIEW

Below is a simplified set of building blocks that will provide a basic platform that will help manage the look-through processes.

The platform should consist of five core modules: Ingestor, Entities, Positions, Analytics, GenerativeAI. Each of these modules may have a corresponding UI component and interacts with the database. Your team is free to decide how to manage the development efforts and is expected to integrate the final work as a team. The outcome should be clearly viewed as a single application.

Your team is free to code in any language(s), use any database, or web framework. You may also use open-source libraries, and UI component libraries that are suitable to improve your application's functionality.

HIGH-LEVEL APPLICATION ARCHITECTURE



Module	Purpose
Ingestor	The Ingestor module manages the ingestion of the external funds dataset provided by each of the external providers. - Each funds dataset is provided monthly to GIC and represents the investment position of the financial instruments that are held by the provider. - For example, if the “Magnum Fund” is an external fund, then its monthly dataset contains the financial instruments managed by the fund as of the end of that month. Simulated data of 10 (ten) funds, with 13 (thirteen) monthly datasets is provided.
Entities	The Entities module manages a store of reference data that represent additional market information about the financial instruments being traded. These datasets may be received from independent data providers (e.g. MSCI, Bloomberg) services on a regular basis. They can be used to enrich the ingested datasets received from the funds. Simulated data of reference data for a set of financial instruments is provided.
Positions	The Positions module manages the detailed positions information about the financial instruments traded by the external fund manager. Simulated data is <u>not</u> provided.
Analytics	The Analytics module manages on-demand analytics services for the application. Simulated data is <u>not</u> provided.
Generative AI	The Generative AI module acts as a proxy between the application and the Anthropic API service. This is to explore how generative ai could be used to address unstructured questions from the management team about the external funds performance and risks. Simulated data is <u>not</u> provided.

EXPECTATIONS

The team should receive the following input files:

Item	Description	File information
1	Master reference tables: • Instrument Reference • Price Reference	master-reference.db
2	External Funds data: • Monthly positions data	e.g. magnum-fund-2022-08-31.csv

The dataset should be available to your team via the following Github repository:

- <https://github.com/gic-codetoimpact/2023-dataset>

In the following pages, your team will find a list of minimal requirements that the application should deliver. These should have the respective REST APIs (or GraphQL) implemented, along with the appropriate error handling mechanisms. Clean architecture and engineering are strongly encouraged.

MODULE: INGESTOR

This module is responsible for managing the ingestion process for the external funds' dataset provided by each of the external providers.

Users are expected to be able to perform the following actions:

#	Requirement
✓ 1.	Develop a process to combine the monthly positions from the external funds dataset into a common positions module
2.	Identify the unique list of instruments from the funds dataset and update the entities module

Explanatory notes:

- Based on the datasets provided, develop basic assumptions to design the process in a simple manner.
- The ingestor module would update the data tables managed by the entities and positions modules.

MODULE: ENTITIES

This module is responsible for managing the list of instruments that have been historically traded by GIC's external funds.

Users are expected to be able to perform the following actions:

#	Requirement
✓ 1.	View the list of instruments
2.	View details of an instrument
3.	View month-end price values of an instrument across time
4.	View the created date/time
5.	View the last modified date/time
6.	Edit an instrument
7.	Edit the "Country" field of a selected instrument
8.	Edit the "Sector" field of a selected instrument

select an instrument
edit details

Explanatory notes:

- New instruments should be automatically added on an ongoing basis based on the ingestion of the monthly funds' dataset (see the Ingestor module). For example, if in the month of January 2023, a particular fund is seen to have an investment holding of a new instrument, then the instrument should be included in the list of instruments in the entities module.
 - The master reference tables can be found in the sqlite database that is provided (master-reference.db). For this challenge you can assume that the master reference tables are read-only and come from an external system. In the real world, the reference tables can contain millions of different instruments across time and will be impractical to replicate the database for this application.
- update db real-time*

DATA SCHEMA

Sample data fields that are required for the instrument reference table

Fields	Description	Example
instrumentId	Primary key of the instrument	1
instrumentName	Name of the instrument	TSY INFL IX N/B 1% 02/15/2048
instrumentType	Type of instrument (Equity, Government Bond)	Government Bond
currency	Currency of the instrument	USD
isinCode	ISIN code, if any	US912810SB52
sedolCode	SEDOL code, if any	BFZBGW6
symbol	Symbol, if any	-
country	Country (US, UK, CN, etc based on ISO 3166-1 Alpha-2 code)	US
sector	Sector	Health Care
createdAt	Datetime created	2023-01-01T00:00:00+0800
modifiedAt	Datetime modified	2023-01-01T00:00:00+0800
... more ...	As deemed necessary by the team	-

Sample data fields that are required for the pricing reference table

Fields	Description	Example
instrumentId	Foreign key of the instrument	1
unitPrice	Price per unit of the instrument (in USD)	107.09
reportedDate	Reported date of the market price	2022-12-31
createdAt	Date created	2023-01-01T00:00:00+0800
modifiedAt	Date modified	2023-01-01T00:00:00+0800
... more ...	If any is deemed necessary by the team	-

RETRIEVE LIST OF INSTRUMENTS

GET /instruments

Retrieve a list of instruments

-

RETRIEVE DETAILS OF AN INSTRUMENT

GET /instruments/{id}

Retrieve an instrument by an id

id	<i>required</i>	Primary key of the instrument	1
----	-----------------	-------------------------------	---

```
{
  "instrumentId": 1,
  "instrumentName": "TSY INFL IX N/B 1% 02/15/2048",
  "instrumentType": "Government Bond",
  "currency": "USD",
  "isinCode": "US912810SB52",
  "sedolCode": "BFZBGW6",
  "symbol": null,
  "country": "US",
  "sector": "Health Care",
  "createdAt": "2023-01-01T00:00:00",
  "modifiedAt": "2023-01-01T00:00:00"
}
```

UPDATE AN INSTRUMENT

PUT /instruments/{id}

Update an instrument.

country	<i>required</i>	Country	US
sector	<i>required</i>	Sector	Health Care
instrumentType	<i>required</i>	Type of instrument (Equity, Government Bond)	Government Bond

RETRIEVE MONTHLY PRICE VALUE RECORDS OF AN INSTRUMENT

GET /price-values/{id}

Get month-end price value records of a particular instrument

id	<i>required</i>	Primary key of the instrument	1
----	-----------------	-------------------------------	---

```
[
  {
    "instrumentId": 1,
    "unitPrice": 10.00,
    "unitPriceDate": "2022-08-31T00:00:00",
    "createdAt": "2023-01-01T00:00:00",
    "modifiedAt": "2023-01-01T00:00:00"
  },
  {
    "instrumentId": 1,
    "unitPrice": 10.03,
    "unitPriceDate": "2022-09-30T00:00:00",
    "createdAt": "2023-01-01T00:00:00",
    "modifiedAt": "2023-01-01T00:00:00"
  },
  {
    "instrumentId": 1,
    "unitPrice": 9.98,
    "unitPriceDate": "2022-10-31T00:00:00",
    "createdAt": "2023-01-01T00:00:00",
    "modifiedAt": "2023-01-01T00:00:00"
  }
]
```

MODULE: POSITIONS

This module is responsible for managing the market values of the funds held by GIC.

Users are expected to be able to perform the following actions:

#	Requirement
1.	Mark the investment positions to market values, using price values from the entities module
2.	View the instrument's market values of a particular fund

DATA SCHEMA

Sample data fields that are required for the positions data table

Fields	Description	Example
fundId	Foreign key of the fund	1
instrumentId	Foreign key of the instrument	1
quantity	Units as per indicated by the fund	340000.00
marketValue	Market value as per reported date (in USD)	34000000.00
realisedProfitLoss	Realised profit or loss for the month (in USD)	100000.00
reportedDate	Reported date of the market valuation	2022-12-31
createdAt	Date created	2023-01-01T00:00:00+0800
modifiedAt	Date modified	2023-01-01T00:00:00+0800
... more ...	<i>If any is deemed necessary by the team</i>	-

Calculation notes:

- quantity of an instrument is based on the quantity of the instrument as provided by the funds' dataset.
- marketValue of an instrument is based on the quantity of the instrument multiplied by the unit price of the instrument as provided by the entities module. For example, if a certain Fund A has an instrument with quantity of 34,000 and a market price of that instrument is US\$100 as of 31 Dec 2022, then the market value of the instrument position for that fund is US\$3,400,000.

Explanatory notes:

- For this challenge, assume that the coupon rates for bonds have already been factored into the pricing values.
- All prices are assumed to be in USD.

REFRESH INVESTMENT POSITIONS OF A FUND

POST /funds/{id}/refresh

data is updated

Refresh the investment positions of a fund

id	<i>required</i>	Primary key of the fund	1
----	-----------------	-------------------------	---

GET INVESTMENT POSITIONS OF A FUND

GET /funds/{id}/instruments/{instrument_id}

Get the investment position of an instrument in a fund

id	<i>required</i>	Primary key of the fund	1
Instrument_id	<i>required</i>	Primary key of the instrument	1

```
[
  {
    "fundId": 1,
    "instrumentId": 1,
    "quantity": 34000.00,
    "marketValue": 3400000.00,
    "realisedProfitLoss": 100000.00,
    "reportedDate": "2022-12-31",
    "createdAt": "2023-08-01T00:00:00",
    "modifiedAt": "2023-08-01T00:00:00"
  },
  {
    "fundId": 1,
    "instrumentId": 1,
    "quantity": 35000.00,
    "marketValue": 3675000.00,
    "realisedProfitLoss": 150000.00,
    "reportedDate": "2023-01-31",
    "createdAt": "2023-08-01T00:00:00",
    "modifiedAt": "2023-08-01T00:00:00"
  },
  {
    "fundId": 1,
    "instrumentId": 1,
    "quantity": 33000.00,
    "marketValue": 3135000.00,
    "realisedProfitLoss": -300000.00,
    "reportedDate": "2023-02-28",
    "createdAt": "2023-08-01T00:00:00",
    "modifiedAt": "2023-08-01T00:00:00"
  }
]
```

MODULE: ANALYTICS

This module is responsible for providing on-demand analytics computations based on filter determined by the front-end user interface.

Users are expected to be able to perform the following actions:

#	Requirement	instruments, country, sector	date
1.	View the breakdown (by instruments, country, sector) of the total market value of a fund as of a specific date		
2.	View total market value of a single fund(s) across a date range	select fund	date range
3.	View the monthly investment return of a fund(s) across a date range		
4.	View the monthly investment return of an instrument(s) (regardless of fund) across a date range		
5.	View top N performing funds by 12-month investment return as of 31 Aug 2023		

CHECK: is it monthly?

Explanatory Notes:

- Your team is given the flexibility to determine the API definition and structure for this module.
- For this challenge, assume that the investment returns calculations are as follows:

12-month investment return of a fund from 1 Sep 2022 to 31 Aug 2023	$\frac{\text{Market value of fund (31 Aug 2023)} - \text{Market value of fund (31 Aug 2022)}}{\text{Market value of fund (31 Aug 2022)}} - 1$ where market value of a fund includes all instruments and cash.
12-month investment return of an instrument from 1 Sep 2022 to 31 Aug 2023	$\frac{[\text{Market value of instrument (31 Aug 2023)} + \text{Realised Profit or Loss of instrument (1 Sep 2022 to 31 Aug 2023)}] - \text{Market value of instrument (31 Aug 2022)}}{\text{Market value of instrument (31 Aug 2022)}} - 1$

MODULE: GENERATIVE AI

This experimental module is responsible for providing generative ai capabilities using Claude 2, the latest version of Anthropic's large language model (LLM) on Amazon Bedrock.

To capture unstructured questions as user inputs, the team can choose to **extend the analytics module to introduce a basic question and answer user interface**. A possible workflow could be as follows:

- End-user sets the **contextual data** on an analytics page based on filter conditions
- End-user keys in a **question** into a textual input
- The front-end sends both the **question & contextual data** to the application back-end, which then engineers a prompt to the Anthropic API.
- Upon receiving the response, the application **displays the properly formatted response** on the front-end to the end-user.

Typical questions could include:

#	Requirement
1.	Given contextual data that contains the returns of the top N funds across time, perform a ranking of the funds after considering the risk due to the funds' volatility .
2.	Given contextual data that contains the instrument breakdown of a particular fund for a specific date, find out which country or sector performed the best .

Fund v

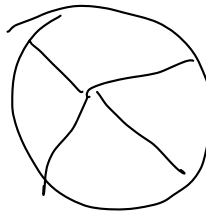
Start | End

Total value

marker
v
:

Instrument name v

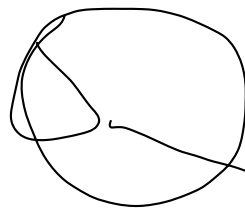
Ins name	Total value	Percentage
----------	-------------	------------



Country v

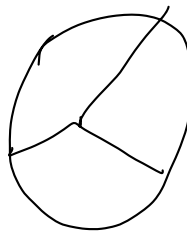
Country	Total val	Percentage
---------	-----------	------------

[LAST PAGE]



Sector v

Sector	Total val	Percentage
--------	-----------	------------



Instrument

Start | End

month 1	_____
month 2	_____



Drinking

EN

